Template Document

Academic Support Agent

February 19, 2025

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1 Introduction

Introduction

- Your introduction goes here!
- Use `itemize` to organize your main points.

Examples Some examples of commonly used commands and features are included, to help you get started.

2 Some LaTeX Examples

2.1 Tables and Figures

Tables and Figures

1. Use `tabular` for basic tables — see Table **??**, for example.
2. You can upload a figure (JPEG, PNG or PDF) using the files menu.
3. To include it in your document, use the `includegraphics` command (see the comment below in the source code).

2.2 Mathematics

Readable Mathematics

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

Slide 1 Goal Main goal that we want to prove.

Table 1: An Example table

Apples	1	2
Oranges	3	4

- Something.
- Something more.
- and more!.

Relaxed Goal relaxation

NAE PCP Verifier from Label Cover using Long Code Reduction

[c] .45 [auto,every node/.style=draw=black,circle,inner sep=0pt,minimum size=1cm] [fill=orange,opacity=.3,draw=white] (0,-4) ellipse (2cm and 0.5cm); [fill=orange,opacity=.3,draw=white] (0,-2) ellipse (2cm and 0.5cm);

[minimum size=0.7cm,draw=white] at (-2.5,-4) (U) U; [minimum size=0.7cm,draw=white] at (-2.5,-2) (V) V;

[minimum size=0.7cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (-1,-4) (nodeU) u; [minimum size=0.7cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (1,-4) (nodeUp)w; [minimum size=0.7cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (0,-4) (nodeUpp);

[minimum size=0.5cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (-1,-2) (nodeVpp); [minimum size=0.5cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (0,-2) (nodeV)v; [minimum size=0.5cm,fill=orange,opacity=.6, text opacity=1,draw=orange] at (1,-2) (nodeVp);

nodeV) edgenode[minimumsize=0cm, draw=white, opacity=0, textopacity=1]{π}nodeU) ; nodeV) edgenode[minimumsize=0cm, draw=white, opacity=0, textopacity=1]{$\pi' $}nodeUp) ;

[auto,every node/.style=draw=white,circle,inner sep=0pt,minimum size=1cm] [rectangle,minimum size=0,inner sep = 0pt,draw=white] at (0,1.5) (B) ; [rectangle,minimum size=0,inner sep = 0pt,draw=white] at (0,1) (A) $A_u : (_2)^{3r}_2 \to_2$;

[rectangle,inner sep = 2pt,fill=green,opacity=.4, text opacity=1,draw=green] at (0,-2) (V) V;

[rectangle,fill=green,opacity=.4, text opacity=1,draw=black] at (-2,0) (Aui) A_{u_i}; [rectangle,fill=green,opacity=.4, text opacity=1,draw=black] at (-1,0) (Auj) A_w; [rectangle,fill=green,opacity=.4, text opacity=1,draw=black] at (0,0) (Auc) A_{u_c}; [rectangle,fill=green,opacity=.4, text opacity=1,draw=black] at (1,0) (Aut) A_{u_t}; [rectangle,fill=green,opacity=.4, text opacity=1,draw=black] at (2,0) (Aus) A_u;

V) edge[->, bendleft] Auj) edge [->] (Auj) edge [->] (Aus) edge [->,bend right] (Aus);

.55

- Query
  1. $A_u(e), A_u(e + f \circ \pi + 1 + \eta)$
  2. $A_w(e'), A_w(e' + f \circ \pi' + \eta')$
- Where
  1. $e, e' :^{3r}_2 \to \{0, 1\}$, $f :^{r}_2 \to \{0, 1\}$
  2. η, η' from noise distribution.
- Correct proofs are Long Code encodings of labels to U given by

$$A_u = (f(a))_{f \in (_2)^{3r}_2}$$

Bottleneck! : Proof size is $2^{2^{3r}} n^r$.

Cannot go beyond $r = O(\log \log n)$.